



Centration Charts

Artisan
LAB NETWORK

HOYA
FOR THE VISIONARIES



Redefining every aspect of vision correction

HOYA Vision Care provides precision laboratory services and high quality ophthalmic lenses to eye care professionals across the country. HOYA invests heavily in research and development. The result is a world-renowned range of products providing total eye care.





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Ensuring a proper frame fit before recording your measurements

Frame selection is as important as the lens you choose. Not only must frames accommodate the lenses, they should also fit comfortably and suit the patient's face shape. Once the frame and lens type have been chosen, the fitting of the lens is very important. Proper fit will ensure less chances of having to redo your job.

Check the frame placement for proper alignment before taking any measurements. If the frame is properly adjusted, the patient will achieve the best vision.

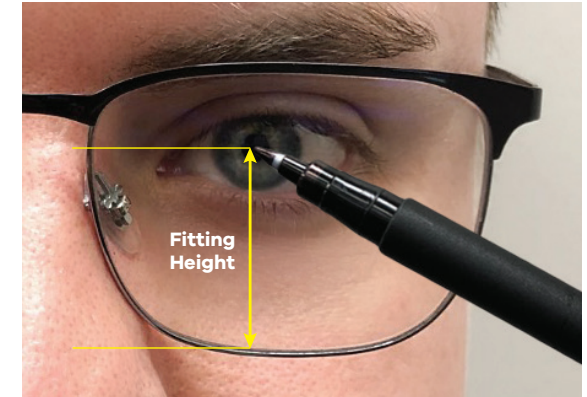
Take initial measurements

- Ensure your patient is looking straight ahead and mark each lens at the pupil center
- Draw a horizontal line on each lens
- Measure fitting height from the deepest point of the lenses to the pupil center
- Take the monocular PD measurements of each eye for far vision using a corneal reflection pupilometer, with the instrument set to infinity (∞)
- **For more accurate results, or to take position of wear (POW) measurements, consider using a digital measuring device**

Check lens cut out

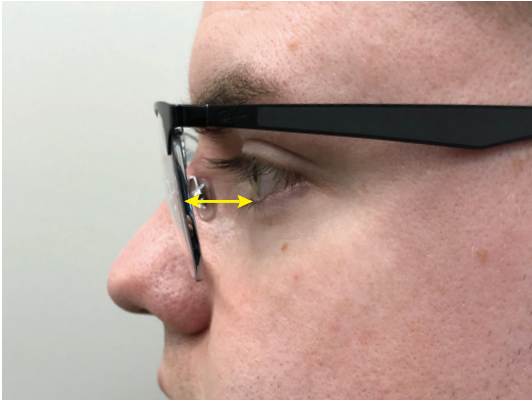
- Select the appropriate centration chart to check the lenses you have chosen
- Use the patient's fitting height and distance PD measurements. Lines should bisect to create a cross (+)
- Place lens cross over the layout chart to verify the lens will fit in the frame

Following the previous steps will ensure a proper fit and provide the best possible vision for your patients.



How to measure fitting heights

With your eyes at the same level as the patient, and your patient's focusing on the bridge of your nose, mark the lens at the pupil center for each eye. Then measure to the deepest point on the lens.

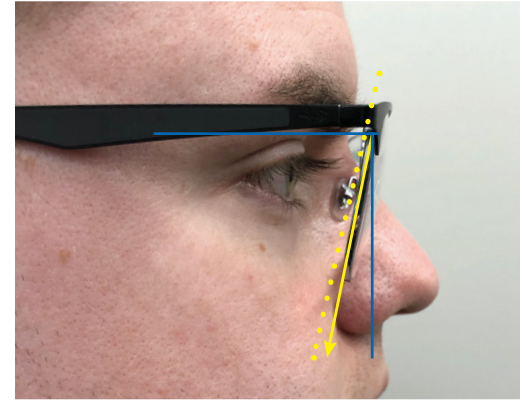


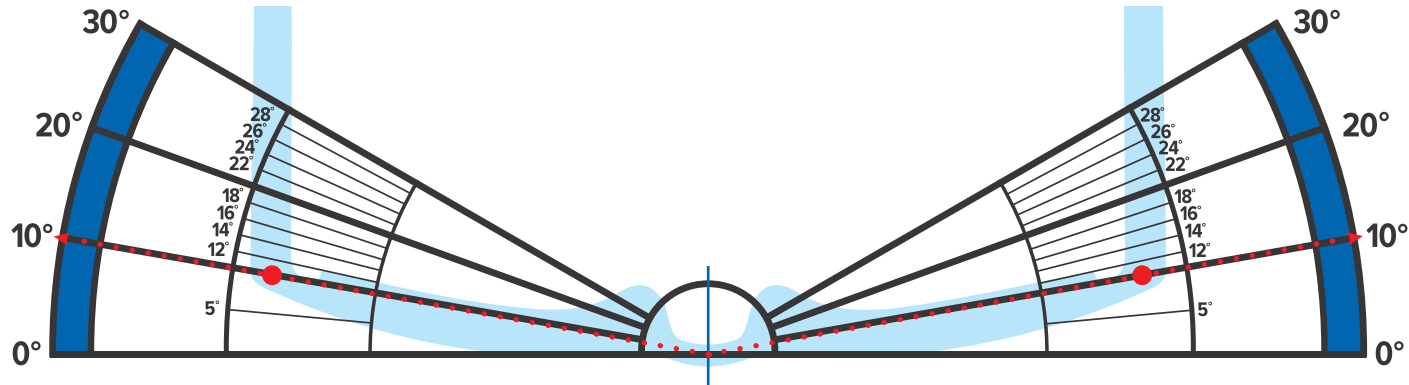
Measuring Vertex Distance (mm)

When wearing glasses, the distance from the front of the cornea to the back surface of the lens is called Vertex Distance. **TIP** - If you can't see the back of the lens, measure from the front of the lens to the front of the cornea and then subtract the lens thickness.

Measuring Pantoscopic Tilt

Pantoscopic tilt refers to the lens frame alignment in the up and down position as it tilts towards the wearers' face. An increased pantoscopic tilt brings the near zone closer to the eye and increases the field of view through the near zone of the lens.





Measuring Frame Wrap Angle

Frame wrap indicates the amount of curvature a frame has towards the patient's face. Frames with more face-form will curve more closely towards the wearer's face, while a frame with less face-form has a flatter frame front.

Frames should have a slight degree of wrap angle. To increase the amount of face-form, use both hands to apply pressure at the bridge and

move it slightly inward. To decrease the amount of face-form, place both hands on the bridge area and gently move the frame front outward.

Adjustments to the face-form angle can help patients with peripheral distortion problems. Try increasing the amount of face-form curvature slightly for patients experiencing discomfort with the peripheral area of their lenses.

Instructions

- Position the frame on the straight edge with temples toward the curved edge
- Center bridge
- Position R & L hinges so that same angle is observed on either side
- Wrap angle in this example is 10.0°



Edge Type Selection



Choosing the correct 'Edge Type' can make the difference between a job being completed on time or being delayed unnecessarily. If you are prompted for an Edge or Frame Type using any of the platforms listed below, this guide will make it easy to correctly identify the required edge type.

Ordering Platforms:

- DVi - RxWizard
- VisionWeb
- Eyefinity
- Direct Integration
- Fax/Phone

Edge Type Options:

- Regular Bevel (Full metal or plastic frames)
- Nylon Groove
- Metal Groove Frame
- Drill Mount Frame with plugs
- Drill Mount Frame with screw
- Specialty Frame requiring notched edging
- Industrial Frames (safety stamp & thickness)*



A **Nylon Groove** rimless frame (GD) with metal/plastic on the top/bottom utilizes a fishing line all around the outside of the lens to hold the lens into place.



Metal Top Groove Frame
with nylon going around
entire lens

Metal Bottom Groove Frame
with nylon going around
entire lens

* Note the fishing line goes around the entire edge of the lens.

* Most grooved rimless frames can be produced to a minimum edge thickness of **2.0mm – 2.2mm**.

To avoid chipping and breakages, we recommend grooving only 1.53 index or higher. 1.50 index (CR39) can be grooved and fitted to a nylon grooving (GD) frame but we offer no chipping warranty due to the fragility of the material.

A **Metal Groove** frame (MG) is where the entire lens is grooved all the way around. By doing so, the lens is secured within the frame without using the nylon fishing line. This means that in order to accommodate a wider groove the edge thickness must be increased. Please note that we do not allow 1.50 index (CR39) in MG frames.



Metal Groove Frame
all the way around

Metal Top Groove Frame
with half nylon line on bottom

Metal groove frames require a minimum edge thickness of 2.5 mm. The metal actually goes inside the groove and is much thicker than fishing line.

IMPORTANT REMINDER - Frames that are half nylon and half metal groove (as pictured above right) should be ordered as metal groove (MG).

* Thickness may vary based on lens material to meet Canadian Safety Standards.



Base Curve Selection using the Simple Base Curve Selection Chart

The base curve is the single front surface of a lens, measured in diopters over the distance portion of a lens. Opticians should consider the patient's current lens material's true surface power and the new prescription, when designing new lenses.

Please note that there are cases where digital, high-power and specialty lenses have limitations and often we cannot control base curve selection. Back side designs in plus Rx may carry a higher than normal base curve, due to the nature of the lens design. Likewise, wrap frames / sunglasses may require a higher base selection.



Power	Recommended Base Curve
+8.25 & Above	8.50 & Above
+6.25 to +8.00	6.50 to 8.25
+4.25 to 6.00	5.00 to 7.00
+0.25 to 4.00	4.00 to 5.00
+0.00 to -4.00	2.50 to 4.25
-4.25 to -6.00	1.25 to 3.25
-6.25 to -8.00	0.00 to 1.25

It is often not possible to deviate from the recommended base curve by more than 2 diopters. E.g. Requesting 8 base on a -5.00 will be problematic.



Laser Engravings

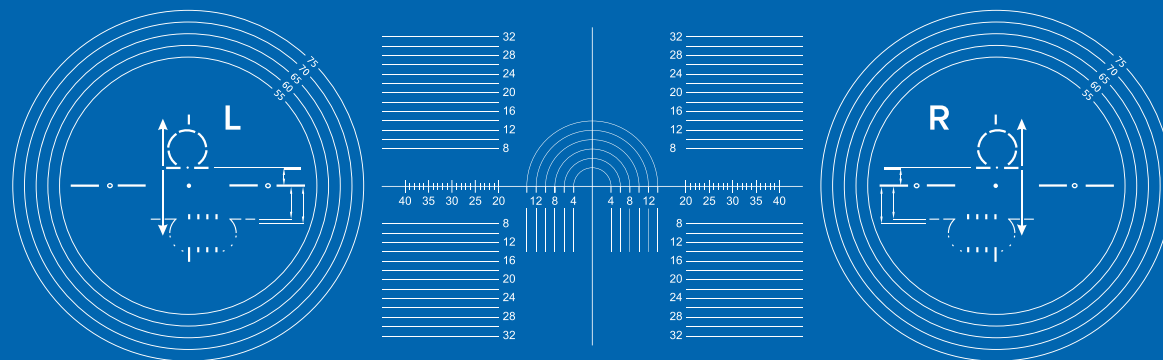
Progressive Lenses		
Lens Style	Minimum Fitting Height	Laser Engraving
iD MyStyle 2	14mm	MSV
iD LifeStyle 3	14mm	L
Array 2	14mm	AA
Array	14mm	Y
Summit ecp BKS	18mm	YSP
Summit cd BKS	14mm	YCD
Amplitude HD3	18mm	AH
Amplitude HD3 Mini	14mm	AHM
Amplitude BKS	19mm	YA
Amplitude Mini BKS	14mm	YAS
GP Wide BKS	18mm	YW

Occupational Lenses		
Lens Style	Minimum Fitting Height	Laser Engraving
iD Space, iD Screen, iD Zoom	18mm	WSV
Tact BKS	18mm	YT

Single Vision Lenses		
Lens Style	Minimum Fitting Height	Laser Engraving
Sync III	14mm	SY
MySV		Ni



HOYA Centration Charts





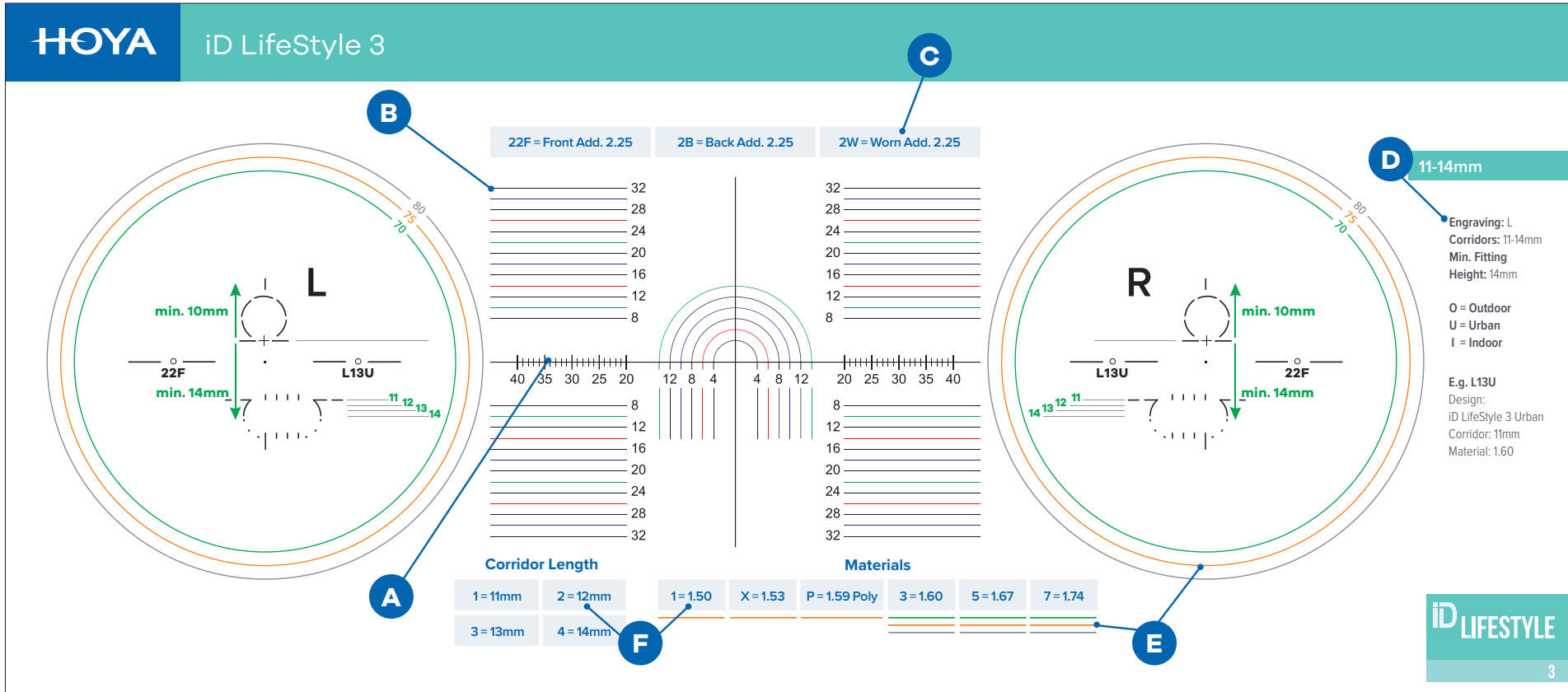
How to read our Centration Charts

Marking your HOYA Lenses

1. Locate and mark your laser engraving points on the temporal and nasal portions of your HOYA lenses using a lens marker.
2. Beginning with your left or right lens, line up the nasal and temporal marks from step 1 on the respective HOYA centration chart. Mark up your center dots.
3. Repeat steps 1 and 2 for the other lens.
4. You can now use these markings to verify your HOYA lenses.

Legend

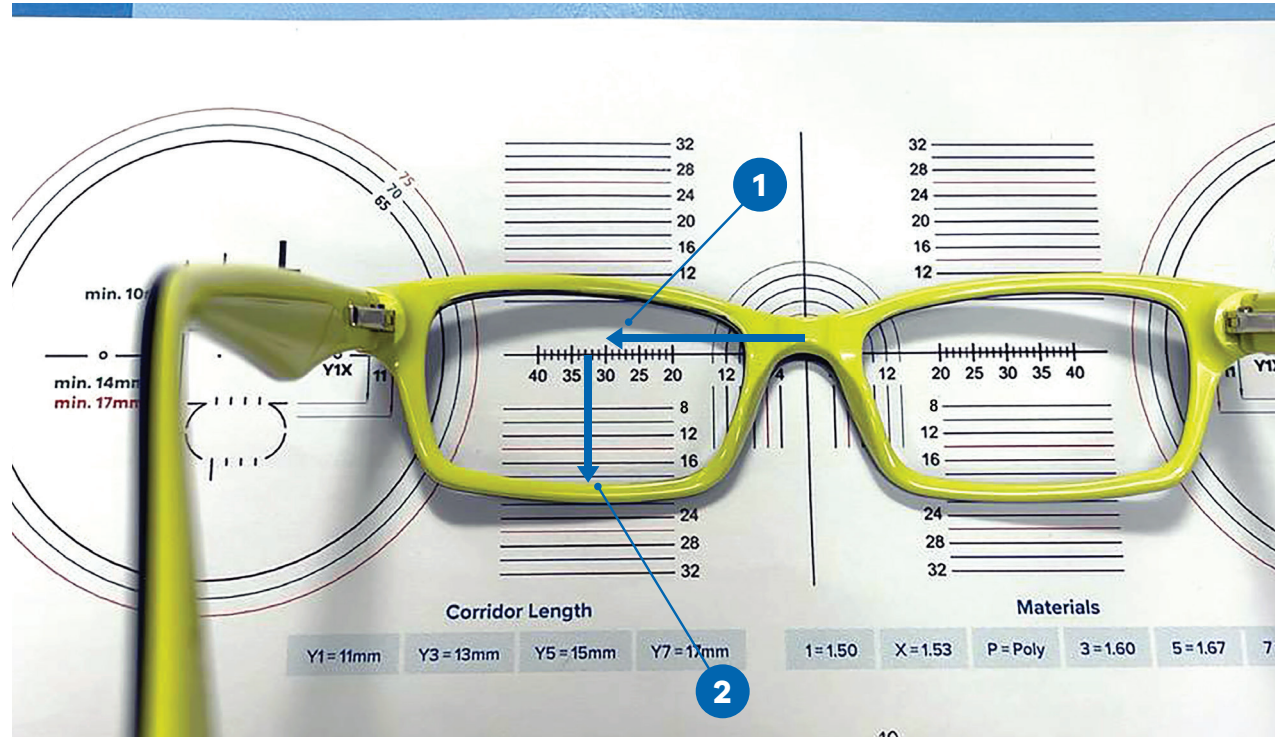
- A** An integrated PD ruler is included on all centration charts. Simply line up the center of the frame bridge on the vertical line in the center and ensure your fitting cross is lined up with the horizontal line or the integrated PD ruler.
- B** The fitting height can be confirmed using the scale below the PD ruler. Simply measure down to the lowest point of the bevel. The top scale can similarly be used for occupational bifocals.
- C** This section of the centration chart will show you engraving details for the add power which will be found on the temporal side of the lens.
- D** This portion of the centration chart will show you the complete laser engraving format to assist in identifying the lens design, material, corridor and add power.
- E** This part of the centration tool can be used to verify cut-out and blank size for your HOYA lenses. These color bars match the lens blank sizes. Diameter is for clear lens only. Some exceptions may apply.
- F** Here you can find the the laser engraving details for corridor lengths and materials.





How to use the center measuring tool

- 1 Pupil Distance
- 2 Fitting Height

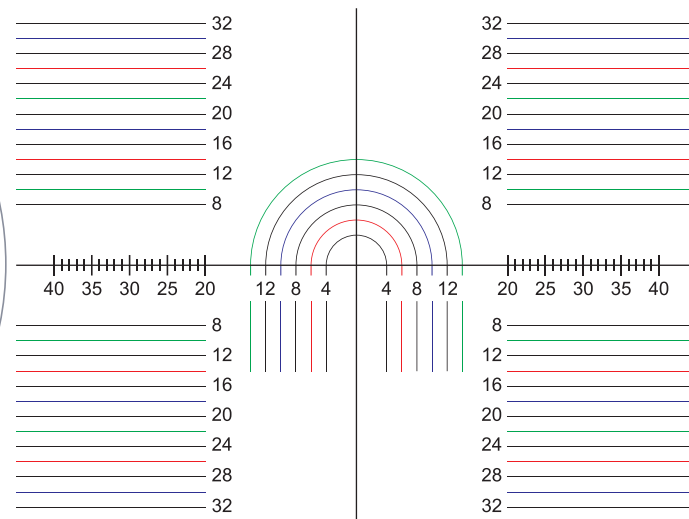
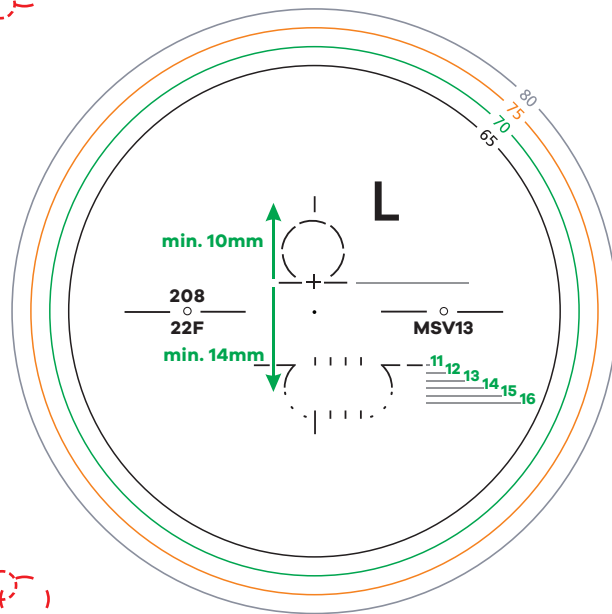


22F = Front Add. 2.25

2B = Back Add. 2.25

2W = Worn Add. 2.25

208 = Design Ref #

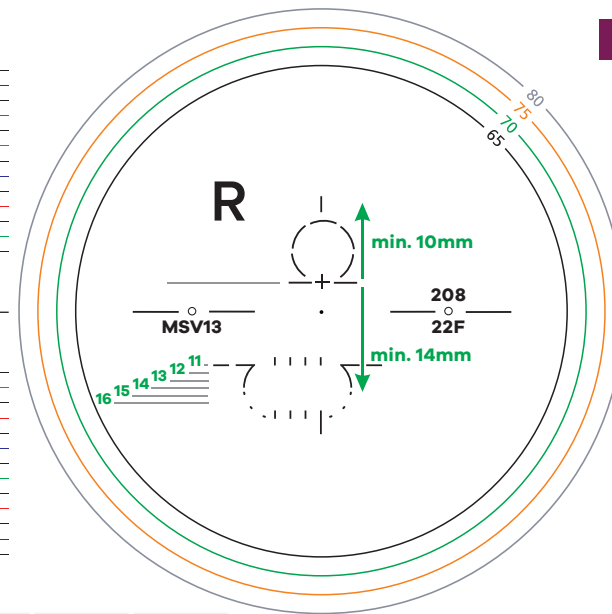


Corridor Length

1 = 11mm	2 = 12mm	3 = 13mm
4 = 14mm	5 = 15mm	6 = 16mm

Materials

1 = 1.50	X = 1.53	P = 1.59 Poly	3 = 1.60	5 = 1.67	7 = 1.74
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11-16mm

Engraving: MSV

Corridors: 11-16mm

Min. Fitting Height:

Personalized to frame

14mm

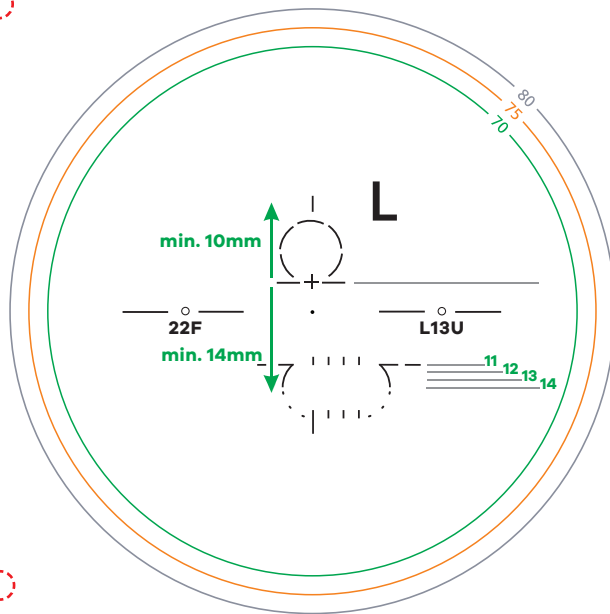
E.g. MSV13

Design: iD MyStyle 2

Corridor: 11mm

Material: 1.60

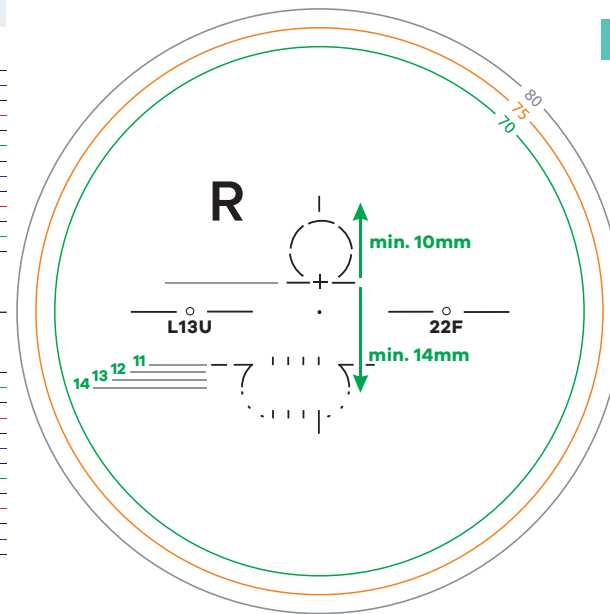
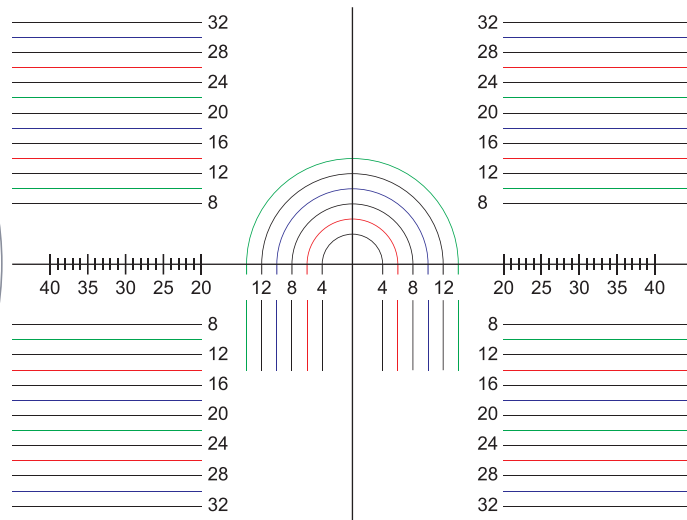
iD MYSTYLE



22F = Front Add. 2.25

2B = Back Add. 2.25

2W = Worn Add. 2.25



11-14mm

Engraving: L
Corridors: 11-14mm
Min. Fitting
Height: 14mm

O = Outdoor
U = Urban
I = Indoor

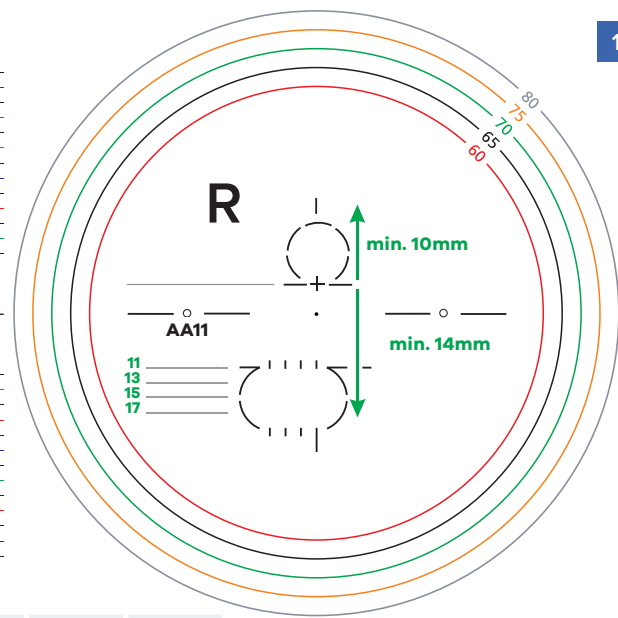
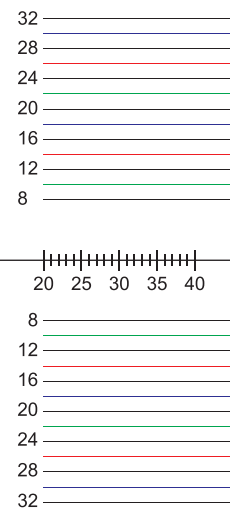
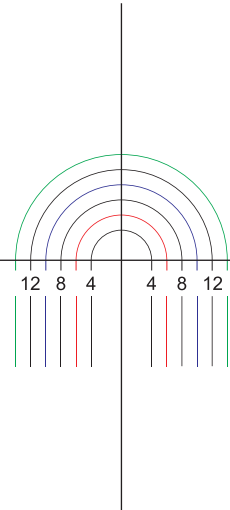
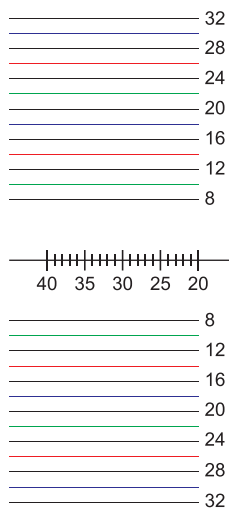
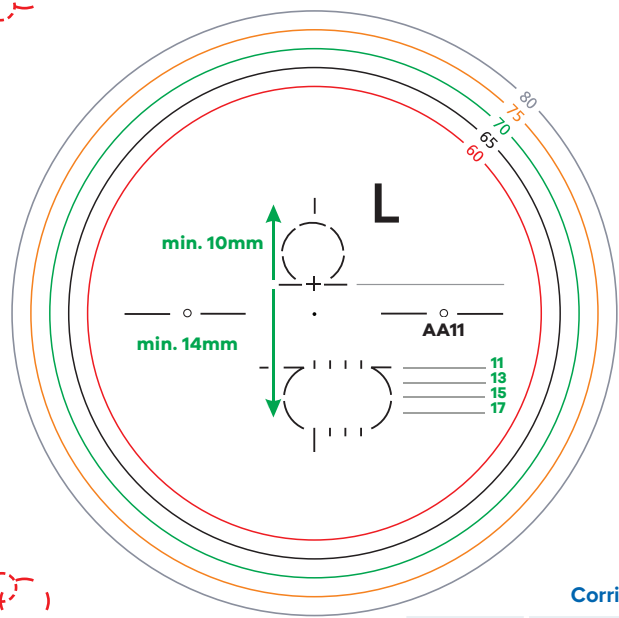
E.g. L13U
Design:
iD LifeStyle 3 Urban
Corridor: 11mm
Material: 1.60

Corridor Length

1 = 11mm	2 = 12mm
3 = 13mm	4 = 14mm

Materials

1 = 1.50	X = 1.53	P = 1.59 Poly	3 = 1.60	5 = 1.67	7 = 1.74
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11, 13, 15, 17mm

Engraving: AA
Corridors: 11, 13, 15, 17mm
Min. Fitting Height: 14mm

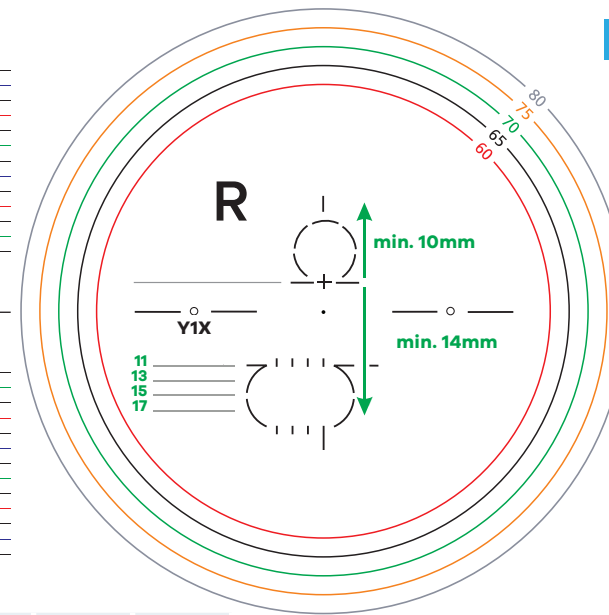
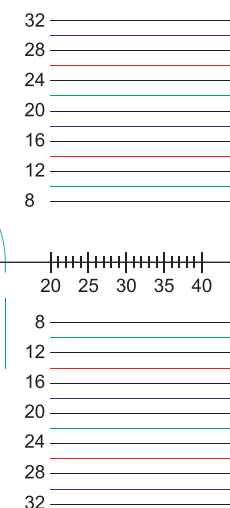
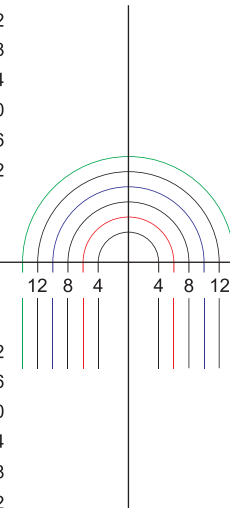
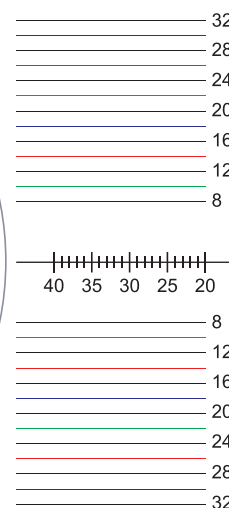
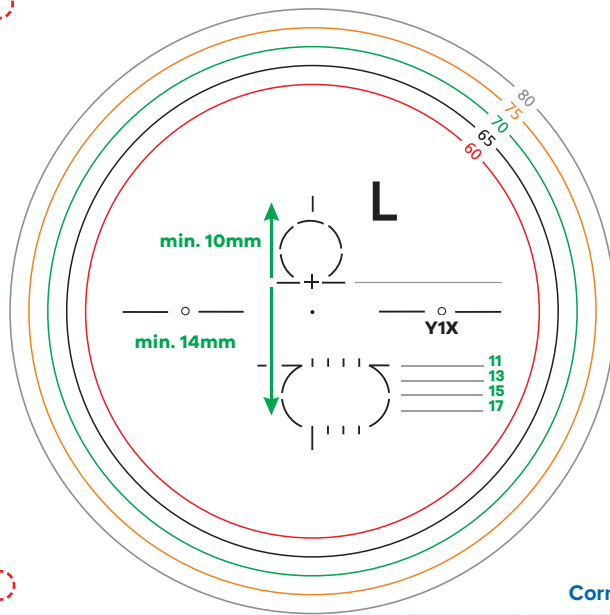
E.g. AA11
Design: Array 2
Corridor: 11mm
Material: 1.50

Corridor Length
AA1 = 11mm AA3 = 13mm AA5 = 15mm AA7 = 17mm

Materials
1 = 1.50 X = 1.53 P = 1.59 Poly 3 = 1.60 5 = 1.67 7 = 1.74

ARRAY

2

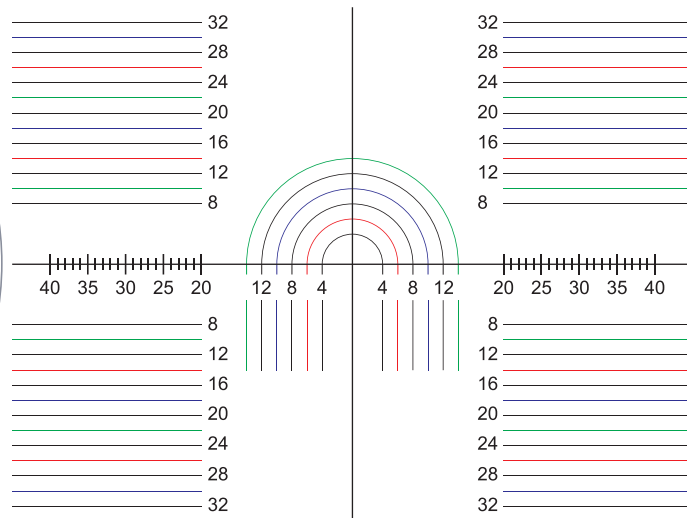
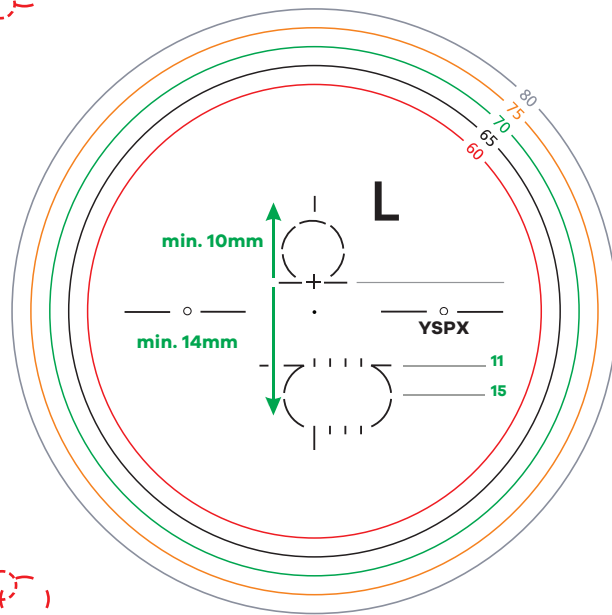


11, 13, 15, 17mm

Engraving: Y
Corridors: 11, 13, 15, 17mm
Min. Fitting Height: 14mm

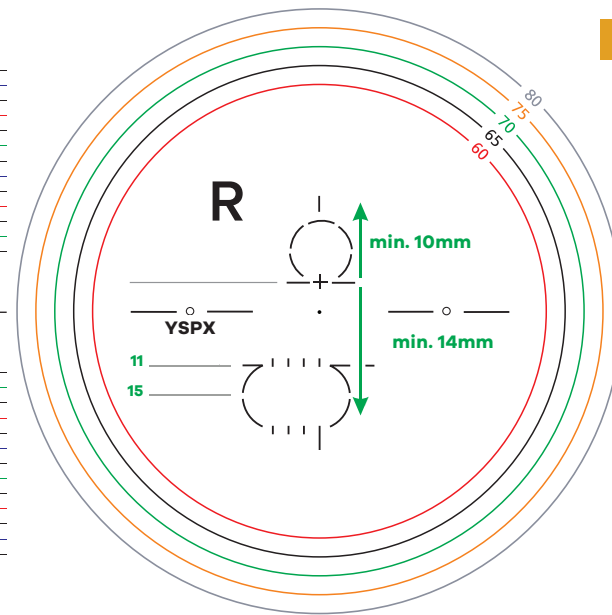
E.g. Y1X
Design: Array
Corridor: 11mm
Material: 1.53

Corridor Length				Materials					
Y1 = 11mm	Y3 = 13mm	Y5 = 15mm	Y7 = 17mm	1 = 1.50	X = 1.53	P = 1.59 Poly	3 = 1.60	5 = 1.67	7 = 1.74



Materials

1 = 1.50	X = 1.53	P = 1.59 Poly	3 = 1.60	5 = 1.67	7 = 1.74
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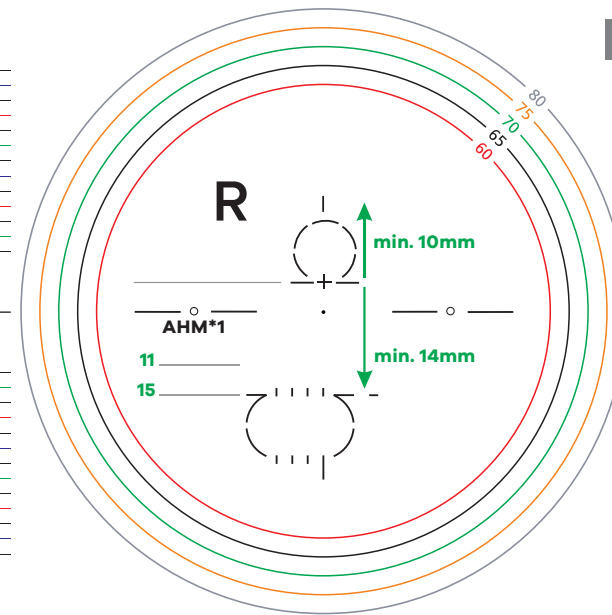
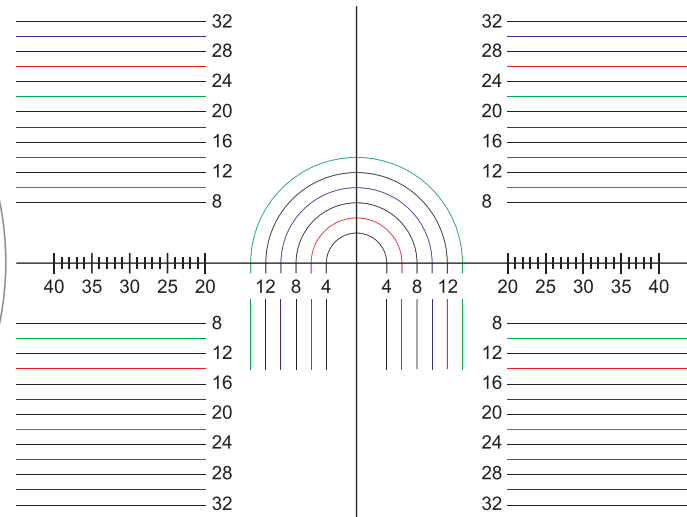
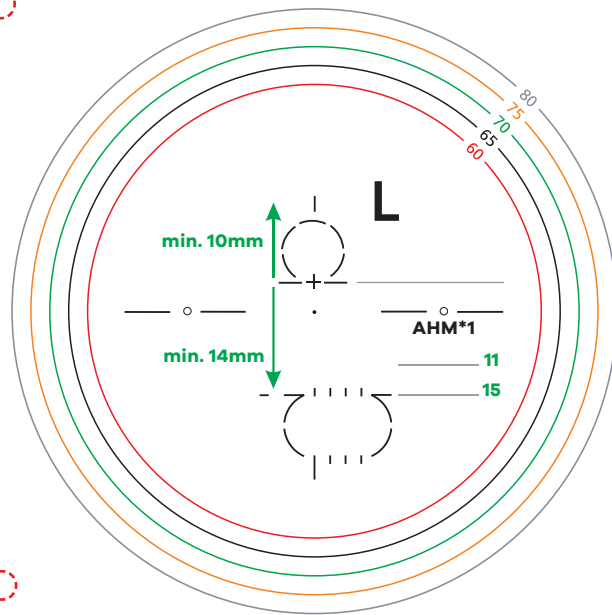
11, 15mm

Engraving: YSP
Corridors: 11, 15mm
Min. Fitting Height: 14mm

E.g. YSPX
Design: Summit ecp BKS
Corridor: 15mm
Material: 1.53

SUMMIT

ECP / CD BKS



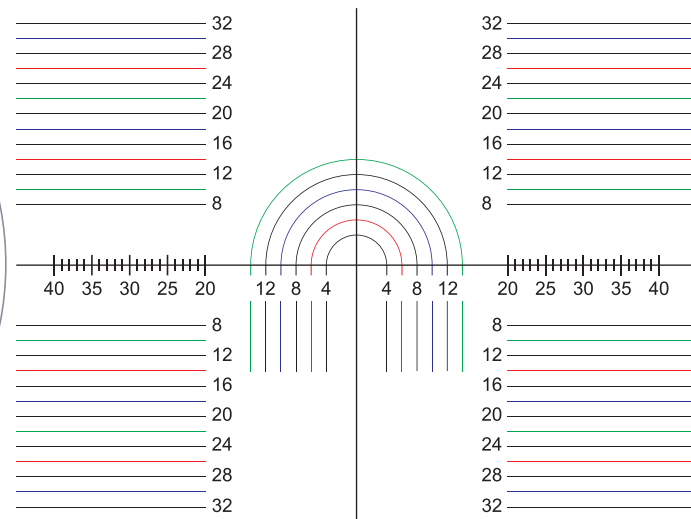
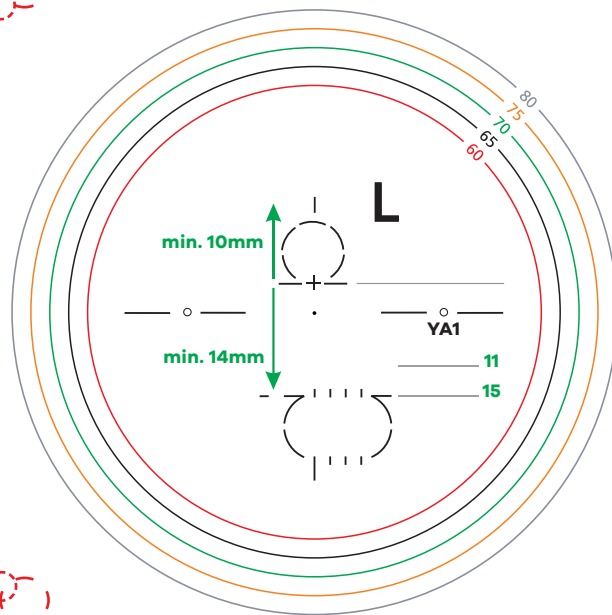
11, 15mm

Engraving: AHM
Corridors: 11, 15mm
Min. Fitting Height: 14mm

E.g. AHM*1
Design: Amplitude HD3
Corridor: 15mm
Material: 1.50

Materials

1=1.50	X=1.53	P=1.59 Poly	3=1.60	5=1.67	7=1.74
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Materials

1 = 1.50	X = 1.53	P = 1.59 Poly	3 = 1.60	5 = 1.67	7 = 1.74
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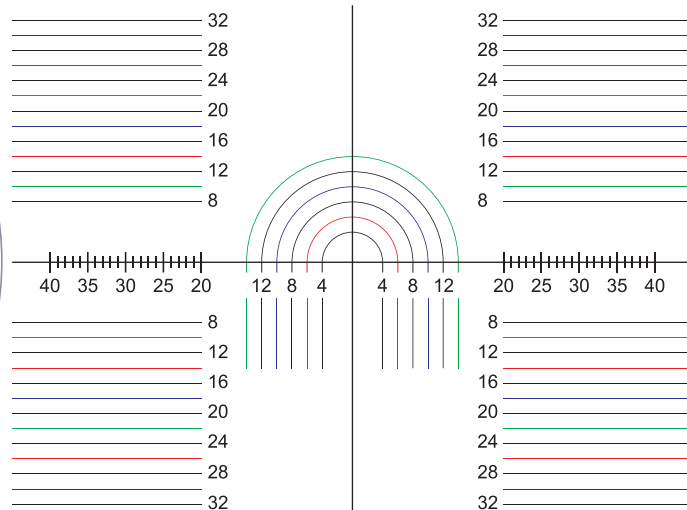
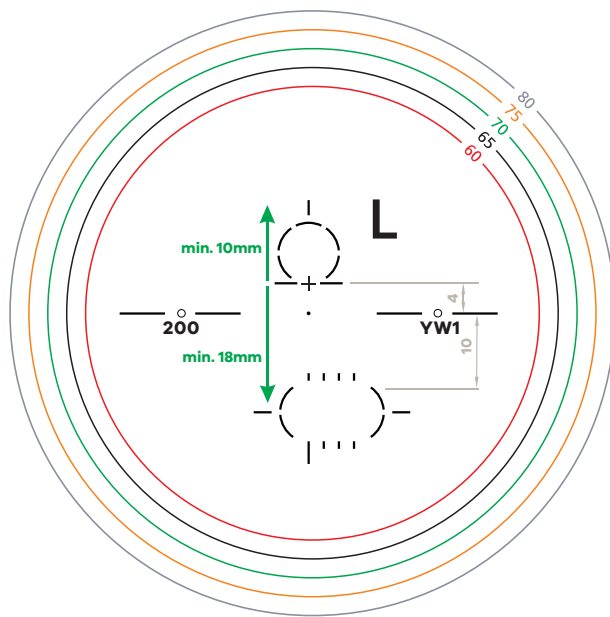
11, 15mm

Engraving: YA
Corridors: 11, 15mm
Min. Fitting Height: 14mm

E.g. YA1
Design: Amplitude BKS
Corridor: 15mm
Material: 1.50

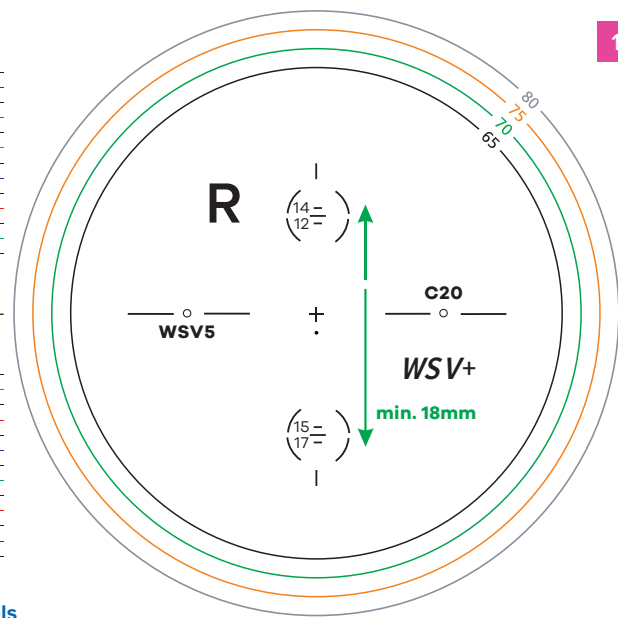
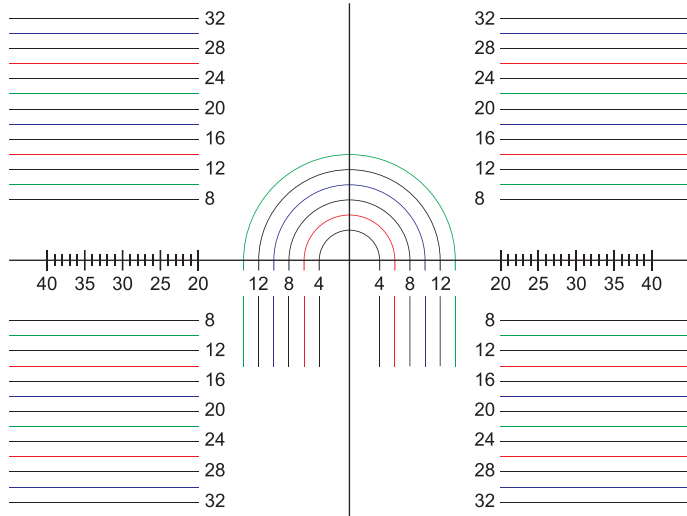
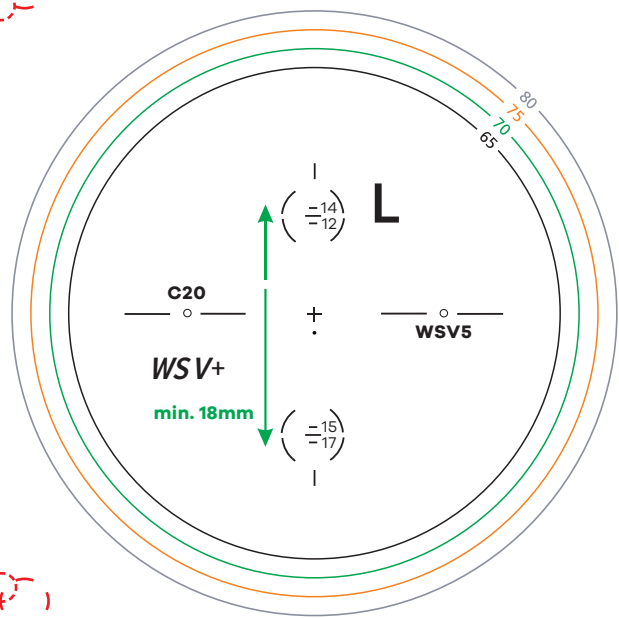
AMPLITUDE

BKS / MINI



Materials

1 = 1.50	X = 1.53	P = 1.59 Poly	3 = 1.60	5 = 1.67	7 = 1.74
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18-24mm

Engraving: WSV
Corridors: 18-24mm
Min. Fitting Height: 18mm*

P = Space
S = Screen
C = Zoom

E.g. WSV5 C20
Design: Zoom
Corridor: 20mm
Material: 1.67

* Recommended fitting height with a frame depth of 28mm min.

Corridor Length (mm)	18	19	20	21	22	23	24
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For Rx verification: the iD Zoom is 50% of the add power at the fitting height (+). The iD Space and iD Screen are 50% of the add power 2.5mm below the fitting height (•)

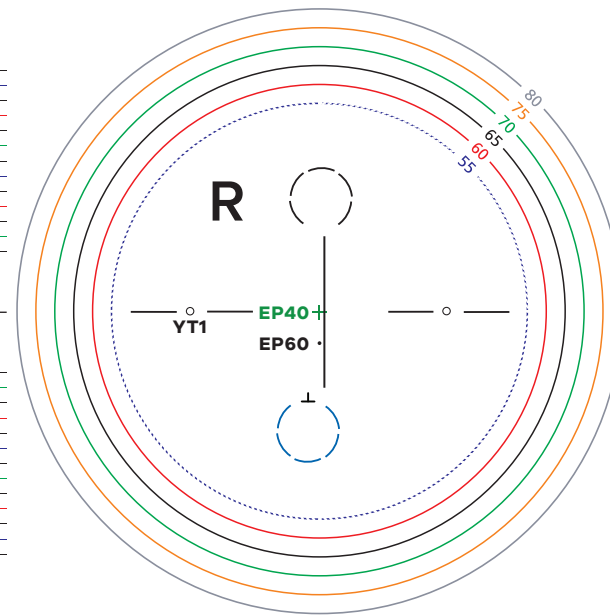
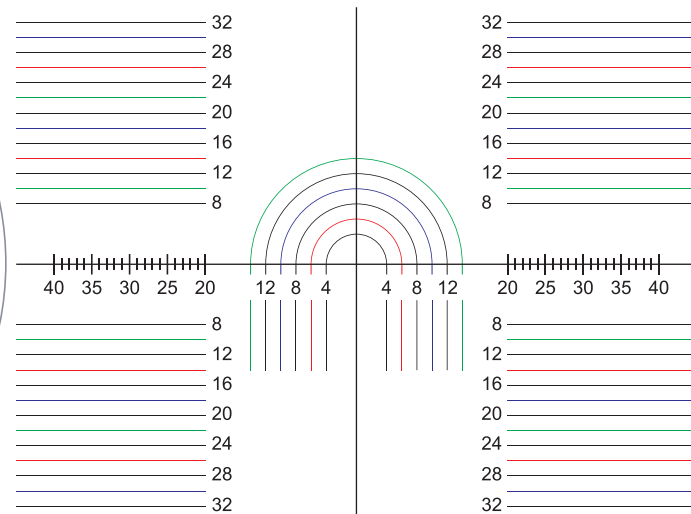
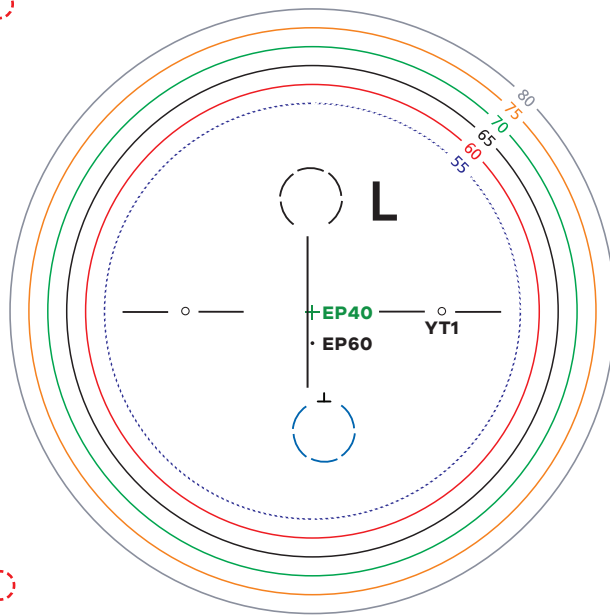
Materials

1=1.50	X=1.53	3=1.60	5=1.67	7=1.74
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iD

NMO

SPACE, SCREEN, ZOOM



Materials

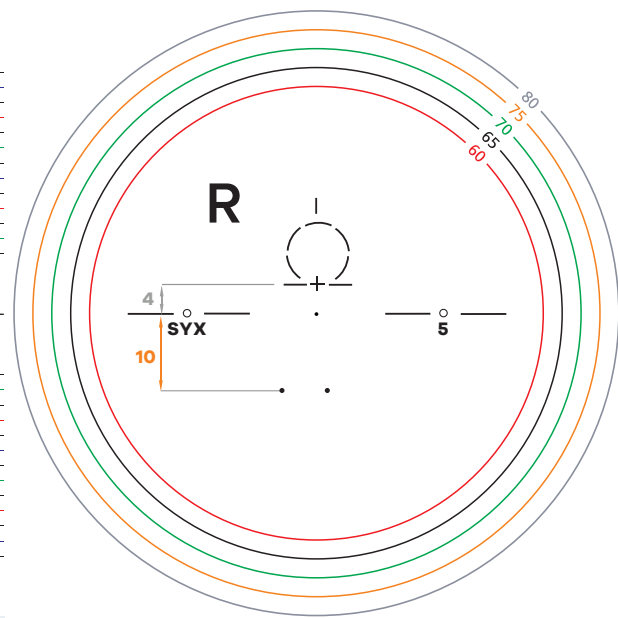
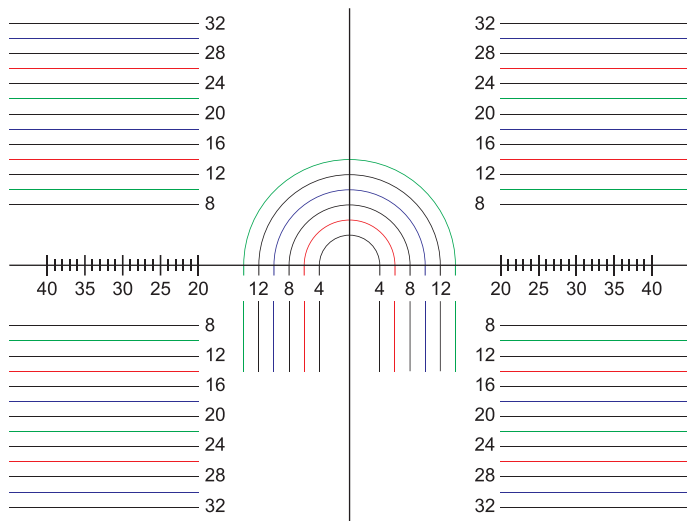
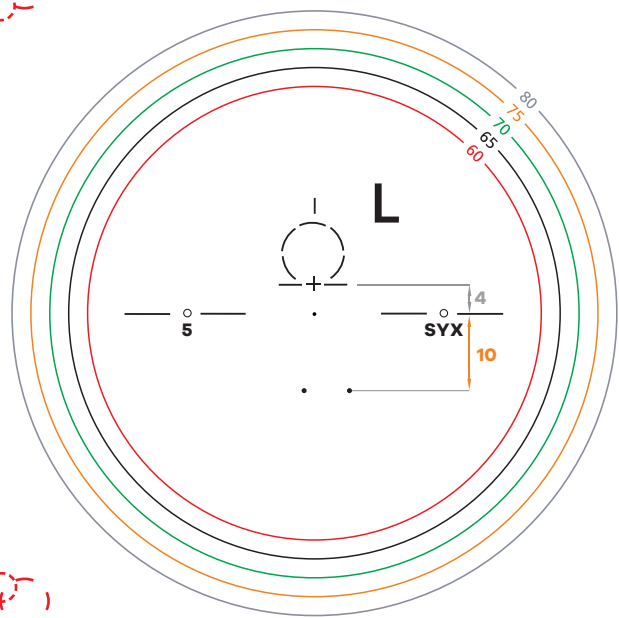
1=1.50	X=1.53	P=1.59 Poly	3=1.60	5=1.67	7=1.74
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Engraving: YT

E.g. YT1
Design: Tact BKS
Material: 1.50

TACT

BKS



Engraving: SY
Corridors: 5/9/13
Min. Fitting Height:
14mm only

E.g. SYX 5
Design: Sync III
Power Boost: 5
Material: 1.53

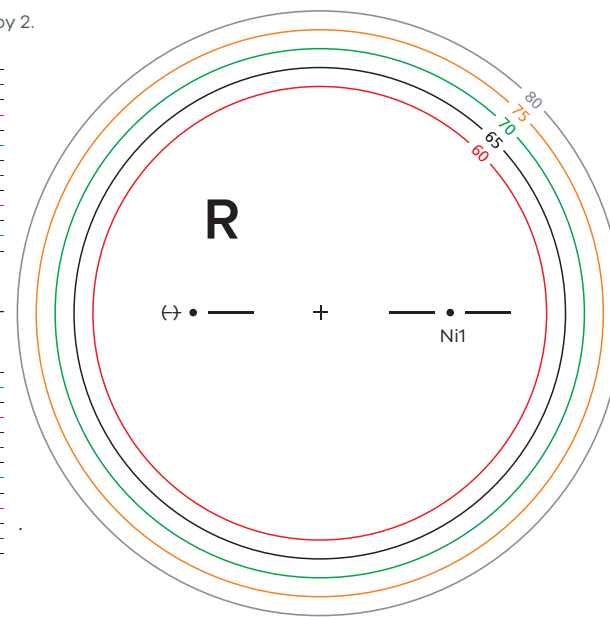
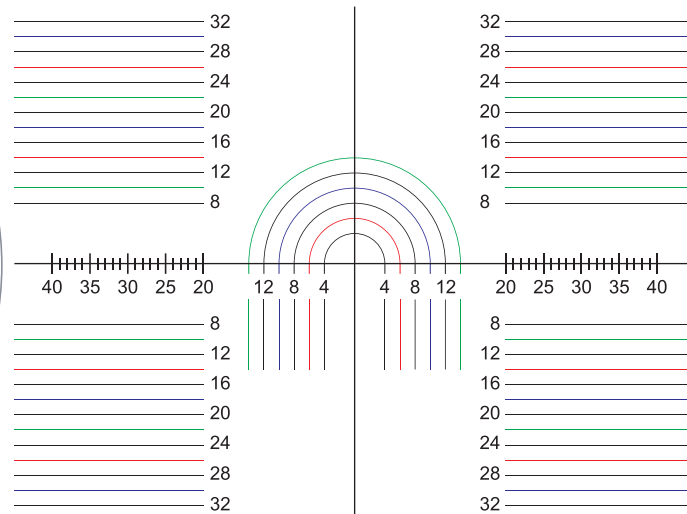
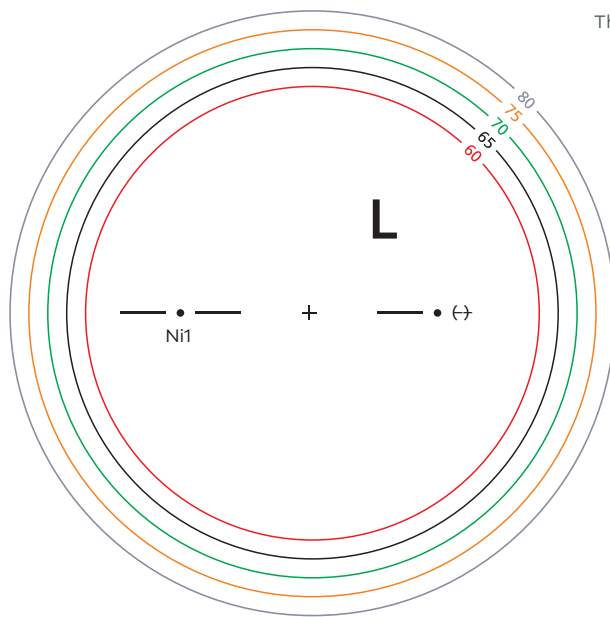
Centering: Center SYNC III lenses according to the monocular distance PD and measure at pupil center when looking straight ahead in horizontal (zero) distance direction.

1=1.50	X=1.53	P=1.59 Poly	3=1.60	5=1.67	7=1.74

Materials

Lens design is normally decentered. Prism reference point will often times not be geometric center of lens.

HOYA strongly recommends including OC height when placing orders.
The default OC is the frame vertical height measurement (B in mm) divided by 2.



Engraving: Ni

E.g. Ni1
Design: MySV
Material: 1.50

Materials

1=1.50	X=1.53	P=1.59 Poly	3=1.60	5=1.67	7=1.74
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**Thank you for choosing
HOYA lenses**

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